AHAMMAD SHAWKI

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EDUCATION

Bangladesh University of Engineering and Technology (BUET) Expected 2028
BSc in Computer Science and Engineering *Dhaka, Bangladesh*
Coursework: Data Structures and Algorithms I and II, Database, Software Engineering, Computer Architecture.

EXPERIENCE

Founder and Developer, SrotDev May 2025 to Present

- Lead an engineering collective shipping production web and mobile applications across React, Next.js, Django, FastAPI and PostgreSQL.
- Architected CareForAll as seven independently deployable Django services behind an API gateway, with its own Prometheus and Grafana observability stack.
- Built Srabon, a gamified science learning platform with multilingual and accessibility support, as the team's flagship product.

President of Web and App Development, NDITC Mar 2022 to Jun 2024

- Led a 30-person engineering team; delivered the rebuilt official website and the UI/UX for the club's native mobile application.
- Raised over 1,000,000 BDT in sponsorship across 100+ company negotiations for INIT, the club's national annual event.
- Taught web development fundamentals directly to more than 500 students as a workshop instructor.

Founder and Team Lead, The AS8 Organization Feb 2021 to May 2022

- Built a developer community that reached more than 20,000 learners through free programmes, courses and technical content.
- Partnered with Codecademy, Hack Club and DeepLearning.AI; wrote articles published on freeCodeCamp News.

PROJECTS

DeepSIFT, auditable digital forensics MCP, Python, Volatility 3, MITRE ATT&CK

- 148-tool Model Context Protocol layer that parses forensic binaries into structured JSON before the model reads them, with a signable chain of custody. Scored 4/4 against published ground truth on two cases, zero hallucinations, 100% claim grounding.

Momentalk, photos that speak for people who cannot SAM 3.1, Llama, TypeScript

- A caregiver photographs a real moment and every person and object in it is outlined and speaks when tapped, so a child who cannot rely on speech points at their own world and it talks for them. Built in the ten days of the Meta Global AI Developer Hackathon, segmentation by SAM 3.1.

Shoulder, private agents that share out family care Strands Agents, A2A protocol, Python

- In three families out of four one adult child becomes the carer, by default. Each sibling gets an agent holding constraints they would never say aloud, and the agents negotiate the rota between themselves. A position can leave an agent; the reason behind it cannot, because the type carrying it has no field for one, a runtime hook checks every message before it leaves, and a wire log audits what crossed. Fair division is deterministic: no model touches those numbers.

MACHINE LEARNING COMPETITIONS

Phantom Road Gaps, Bias Bounty Mapping Equity Humane Intelligence and Zindi, 2026

- A highway on a tract boundary is counted in both tracts, and one carrying two route numbers is counted twice again. Those two rules make up 72% of the mean road coverage gap and bend the equity scorecard built to reveal inequity: counted once, the rural to urban ratio moves from 2.34 to 2.73.

CADENCE, voice-based Parkinson's screening ML Empowerment Build Challenge, 2026

- Won best overall project. Screens from a 30-second voice sample and explains every score, validated across three independent language corpora using cross-database testing and domain-adversarial training to avoid microphone bias.

AWARDS

Fifth place, Top 10 finalist, Data Privacy Avengers 2026, Robi Axiata, with team TheCypherGirls. **Grand Prize Winner**, FLIR App Challenge 2025 to 2026, for SolarSnap. **Winner, Golden Agent**, Everyday Agents category, Agents for Humans Hackathon, 2026, for Shoulder. **Best Overall Project**, ML Empowerment Build Challenge 2.0, 2026, for CADENCE. **Best ERNIE Multimodal Application**, ERNIE AI Developer Challenge by Baidu, 2026, for Doclyst. **Honourable Mention**, LMA EDGE Hackathon, 2026, presented at the London finale, for Coven.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Java, SQL, Dart, HTML/CSS

Frameworks: React, Next.js, Django, Django REST Framework, FastAPI, Flask, Node.js, Flutter, JavaFX

AI and Machine Learning: PyTorch, scikit-learn, Model Context Protocol, Google ADK, Retrieval Augmented Generation, Whisper, Gemini, Claude, Pandas, NumPy

Infrastructure: Docker, Google Cloud (Cloud Run, Pub/Sub, Firestore, Vertex AI), AWS (Bedrock, ECS, Lambda), PostgreSQL, MongoDB, Redis, GitHub Actions, Grafana